DAWL / TECHNICAL HANDOFF

The Hypertrix

Hybrid architecture and experimental results

Technical handoff for Yngvar | Experimental work by Espen | 14 October 2026

The Hypertrix is the working name for the DAWL platform: an open, distributed system intended to connect contributed data, reproducible transformations and, eventually, model weights through verifiable lineage. This report covers the executed hybrid testnet and its local GPU extension. It is an engineering pilot report, not a completed whitepaper or a claim of research novelty.

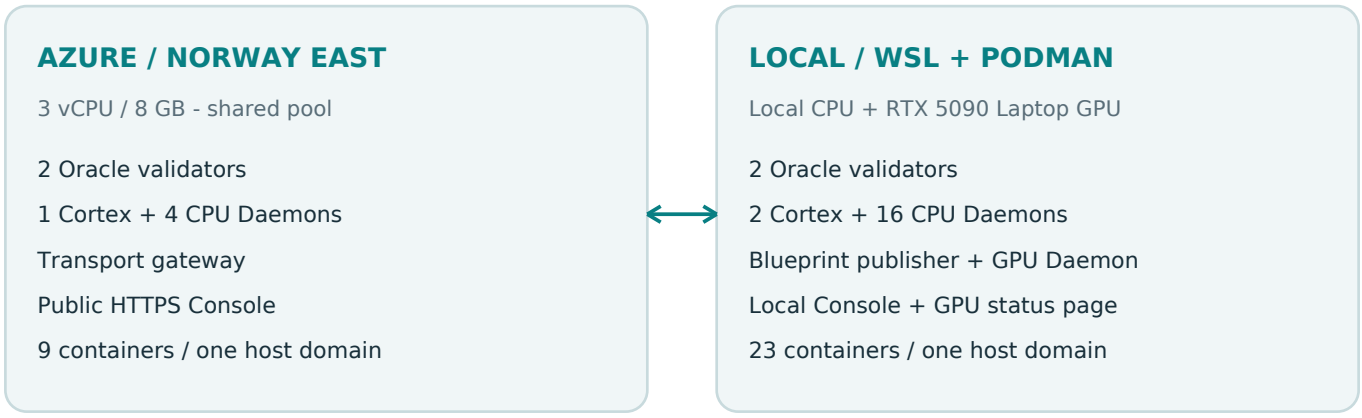
The prototype now joins a local WSL/Podman host to Azure Container Instances in Norway East. Separate node containers ingest and serve data, replay transformations, and commit accepted claims through a four-validator CometBFT network. A newcomer added five books without replacing the original chain; an independent GPU node then reconstructed the entire released corpus.

Current result	Observed value
Released corpus	13 books; 1,807 Frags; 2,061,330 tokens
Accepted consensus audit coverage	34 jobs; 68 receipts; 3,614 Frag replays
Independent GPU diagnostic	All 1,807 Frags matched CPU bytes and ledger hashes
Compact transfer and local reconstruction	172.5 MB fetched; 6.33 GB reconstructed and verified in 66.6 s
Cloud allocation	3 vCPU / 8 GB, nine containers
Network membership	29 node containers; two host failure domains; one operator
Model training and human-participant studies	Neither performed

Main finding. The bounded ingestion-to-release mechanism works across the two sites: content-addressed data, signed provenance, deterministic vector lookup, asynchronous audits and certified release state agree. The evidence supports continued systems research. It does not establish permissionless security, production scalability, beneficial trust effects or the training history of the imported embedding model. (Pooled hybrid evidence; Per-node hybrid evidence; Newcomer contribution evidence; GPU reconstruction evidence)

Evidence covers 13-14 October 2026 in Oslo (UTC+2). The GPU report was signed at 01:38:01 Oslo on 14 October. Measurements remain snapshots from a continuing pilot; a completed long-duration availability study is not claimed.

1. Architecture and deployment



Hypertrix term	Implemented responsibility
The Collective	Signed role discovery and peer artifact transfer over an invited mesh
The Oracle	Four fixed validators order and certify deterministic application state
Cortex Units	Store source/token artifacts and the frozen embedding matrix
Daemons	CPU replay under execution leases; separate GPU diagnostic replay
Cipher-Logs	Signed lineage records and audit claims with explicit scope
The Blueprint	Immutable dataset release, with parent release and input manifests

There are 29 node containers plus one transport gateway and two Console containers: 32 containers in this pilot topology. Same-role nodes reuse an image but have separate keys and state. The GPU container has a 2-CPU/2-GiB limit; local CPU resources are additional to the cloud pool. The GPU shares the physical local host and does not add a failure domain. (Per-node hybrid evidence; Newcomer contribution evidence; GPU reconstruction evidence)

Azure's nine-container group shares networking, scheduling and lifecycle. Two SSH connections separate consensus/control traffic from bulk artifacts. A third, narrow connection was added for the GPU discovery endpoint. Public ingress is SSH 2222 and Console HTTP/HTTPS 80/443. Validator application adapters use private Unix sockets. No validator key is mounted into a provider, gateway or Console. This is process/state separation within a trusted-operator pilot, not a hardened multi-tenant boundary. (Per-node hybrid evidence; Network recovery evidence; Azure container-group reference)

The following two landscape sheets expand this topology into a complete role, protocol and data-flow map. Solid paths show the implemented prototype; dashed amber notes show candidate hardening and future training/weight promotion. Protocol IDs P0-P9 connect the role diagram to the wire/interface inventory. The security notes identify residual risks, not completed mitigations. (Implementation and protocol notes)

Node roles, artifact flow and ledger decisions

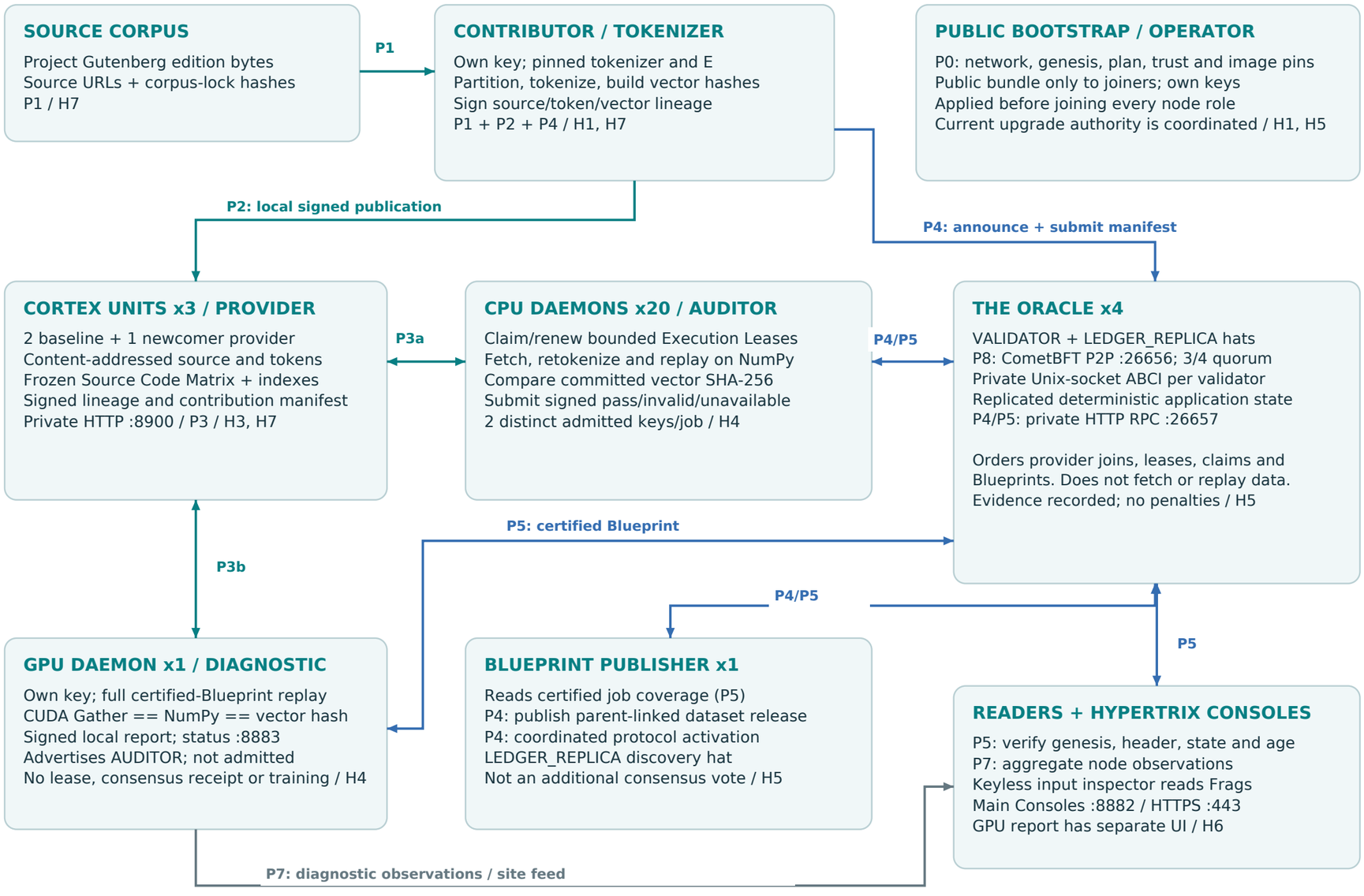
Logical responsibilities are shown separately; the newcomer combines ingestion/tokenization and one of the three Cortex providers in one container.

artifact / local publication

signed command / certified state

observation (not consensus)

P0-P9: protocol inventory on the next sheet.



THE COLLECTIVE / P6 DISCOVERY ON EVERY NODE

Distributed Kademlia-style HTTP :8903. Signed role/endpoint records, XOR routing, expiry and sequence checks. Lookup does not authorize a role. H1/H2.

FUTURE PATH / NOT EXECUTED IN THIS PROTOTYPE

Certified Blueprint + parent weights -> training node -> checkpoint validation / evaluations -> weight-ledger promotion. H8.

H1 / Role admission and identity

P0/P4: scoped, expiring role credentials; revocation. CA-issued or network-approved certificates are alternatives. A key is not a unique operator.

H2 / Eclipse, Sybil and discovery DoS

P6: diverse independent seeds and peer selection; per-peer request/storage quotas. Signed records alone do not make routing Sybil-resistant.

H3 / Data availability under churn

P3: replicas on independent hosts; repair and unpredictable retrieval checks. Hashes detect bad bytes; they do not make a missing provider available.

H4 / Collusion and dummy approvals

P4: post-commit unpredictable challenges, independent reassignment, canaries and dispute review. Two signatures do not prove honest computation.

H5 / Consensus and upgrade authority

P8/P0: independent validator operators, protected remote signers with fencing, threshold-reviewed releases and recovery drills. No automatic slashing now.

H6 / Transport, host and telemetry

P9/P7: peer-bound TLS/Noise or managed mTLS; workload ACLs and isolation; quotas and signed telemetry where needed. Transport auth does not grant a role.

H7 / Poisoning and source governance

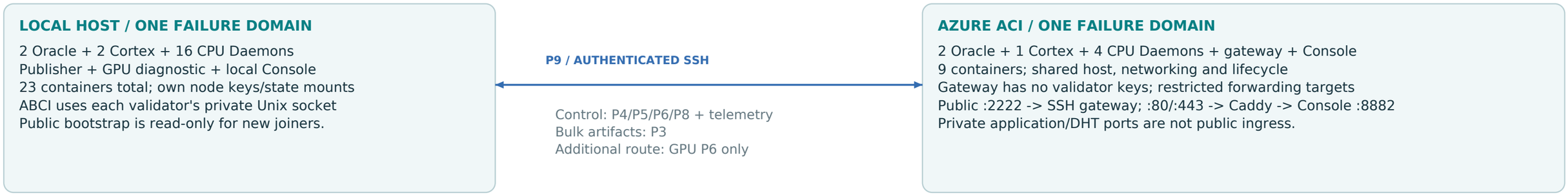
P1/P2: rights/consent policy, deduplication, quarantine and corpus-quality review. Provenance binds origin and bytes, not truth or suitability.

H8 / Future training and weights

Future: bind dataset, parent weights, recipe and environment; validate checkpoints and evaluations before promotion. Exact Gather is not training reproducibility.

Transport boundaries and protocol inventory

P0-P9 identify the actual paths on the role diagram. All amber controls remain candidates; they have not been enabled by this documentation change.



CURRENT TRUST BOUNDARY: one operator controls both sites, bootstrap and upgrade activation. Containers separate state; they do not create independent operators.

ID	Interface	Sender / receiver / transport	Payload, checks and important limits
P0	Bootstrap / runtime trust	Public files -> every role; operator controls issuance	SHA-256 network/genesis/plan/image pins; Ed25519 registrations. Own private state per node. Runtime upgrade is publisher-signed, not threshold governance.
P1	Source acquisition	External corpus -> contributor; HTTPS	Original edition bytes + URL/lock. Content hashing records what was fetched; HTTPS and lineage do not establish factual quality or rights.
P2	Ingestion / local provenance	Contributor -> its Cortex object store; local calls/files	Pinned tokenizer; uint32-le tokens; E[t] float32-le; SHA-256 artifact descriptors; Ed25519 signed parent DAG and author sequence. Dense vectors discarded.
P3a / P3b	Artifact pull / data plane	Private HTTP :8900 -> requester caches	CPU Puller: 1-MiB pieces, deduplication, bounded concurrency, retry/failover, final whole-object hash. GPU/public reader: bounded whole objects. Frozen E is transferred once per empty cache.
P4	Signed commands / leases	Contributor, admitted CPU Daemons and publisher -> Oracle RPC :26657	Canonical JSON + Ed25519; chain/plan binding, actor sequence and validity height. Announce/submit, claim/renew/result and publish. GPU diagnostic does not send these transactions.
P5	Certified reads	Oracle RPC :26657 -> CPU/GPU nodes, publisher, readers and Consoles	Go/CometBFT verifies the fixed-genesis quorum header and state app hash. Minimum accepted height + age window guard reads. Blueprint manifests then bind artifact fetches.
P6	Role discovery / peer routing	Each peer <-> peers/seeds; HTTP :8903	160-bit XOR distance; k=8, alpha=3; signed endpoint/role records, sequence and expiry, lifetime <=180 s. Announcements refresh every 25 s. Invited loopback routes, not open Internet discovery.
P7	Observability / public inspection	Node JSON files/site HTTP -> Console; browser HTTPS :443	Private cloud feed :8901, local feed forwarded :8902; local Console :8882 and GPU page :8883. Certificates are checked separately from unsigned telemetry. Read-only UI; no runtime socket.
P8	Consensus / application execution	Oracle <-> Oracle on TCP :26656; private ABCI Unix sockets	CometBFT fixed voting set; 3 of 4 votes. Per-validator deterministic state and persistent signer state. Incident/conflicting-result evidence can be recorded; no automatic financial punishment.
P9	Cross-host pilot transport	Local outbound SSH -> Azure gateway TCP :2222	Pinned SSH host key and operator authentication; allowlisted forwards. Separate control and bulk-data connections plus GPU DHT route. HTTP is private inside this mesh; no mesh-wide peer mTLS yet.

ROLE SCOPE: VALIDATOR, PROVIDER, TOKENIZER, AUDITOR and LEDGER_REPLICA are advertised hats. Actual audit/vote authority comes from admission rules; the GPU only advertises AUDITOR. TRAINER_GPU is a planned execution capability, not a running training node.

Implementation references: client.py, state.py, incidents.py, worker.py, dht.py, transfer.py, contributor_node.py, gpu_reconstruction_node.py and hybrid_observer.py. Security rationale and validation criteria: docs/architecture_notes.md.

2. Data flow and mathematical contract



Signed parents and content hashes connect the stages; dense vectors are reconstructed locally.

For a source chunk x, pinned tokenizer T and frozen input-embedding matrix E:

```
t = T(x); E in float32[V, d]; X[i, j] = E[t[i], j]
V = 50,257; d = 768; t is uint32 little-endian
c_tokens = SHA256(LE32(t))
c_vectors = SHA256(row_major_float32_le(X))
```

Each Frag is a bounded source/token chunk. The Synapses in this experiment are input-table rows, without positions, contextual layers, pooling or training. Gather performs no floating-point arithmetic. The GPU implementation copies 32-bit words, so signed zero and subnormal bit patterns can survive exactly. Bit-exactness here cannot be generalized to distributed floating-point training.

For L tokens, dense vector bytes equal 4Ld. The current corpus therefore represents 4 x 2,061,330 x 768 = **6,332,405,760 bytes**. The shared matrix is 4Vd = 154,389,504 bytes, and token IDs occupy 4L = 8,245,320 bytes before source and metadata. Reconstruction trades repeated dense-vector storage and transport for a shared table, compact tokens and local computation. It does not eliminate the need to download and retain that table. (GPU reconstruction evidence)

Artifact descriptors bind both byte length and SHA-256. Signed lineage entries name parent IDs, signer, author sequence and predecessor. The local lineage checkpoint hashes a canonical sorted set of entry IDs; it is not described here as a Merkle inclusion proof. Corpus locks bind the exact acquired edition bytes; registries bind their deterministic partitioning. Replay reconstructs original parts, checks token snapshots, retokenizes and compares vector commitments.

The source corpus uses Project Gutenberg texts: original IDs 11, 76, 84, 345, 1342, 1400, 1661 and 2701; added IDs 35, 36, 62, 201 and 5230. Provenance records source URLs and hashes; a signature alone is not a rights or factual-accuracy determination. The handoff contains metadata and evidence, not book bodies or the embedding/model weights. (Per-node hybrid evidence; Newcomer contribution evidence)

3. Trust, consensus and discovery

Mechanism	What a successful check establishes	What it does not establish
SHA-256 and exact size	Retrieved bytes match a commitment	Truth or availability of the source
Ed25519 signature	The claimed key signed the bound bytes	A unique person or honest execution
Independent replay	The checked transformation reproduces its commitment	Quality of future model outputs
Quorum certificate	State bytes are certified under pinned validator policy	Correctness of every worker's computation
DHT role record	A key advertised an endpoint/role for a bounded lifetime	Authorization, capacity or Sybil resistance

Consensus. With four equal-weight validators, q = floor(2n/3) + 1 = 3 votes are required. Two three-vote quorums intersect in at least 2q - n = 2 validators. Under the protocol assumptions, fewer than one-third Byzantine voting power and honest signing behavior prevent conflicting finalization. Liveness needs enough reachable voting power and eventual network synchrony. A 2/2 site split therefore halts both halves; one unavailable validator leaves three able to progress. These are protocol properties, not evidence of four independent operators. Observed fault tests are described in Section 4. (Per-node hybrid evidence; CometBFT implementation reference)

Keyless clients use the Go/CometBFT verifier, a pinned genesis, the signed header and application-state hash, a minimum accepted height and a freshness window. Unverified RPC responses do not silently substitute for a missing certificate. The current validator set is fixed; the coordinated runtime upgrade is publisher-driven and does not demonstrate decentralized upgrade governance.

Discovery. Peer IDs and role lookup keys use 160-bit hashes; routing compares D(a,b) = integer(a XOR b). The bounded Kademlia-style implementation uses bucket size 8, up to 3 parallel requests, and signed advertisements with lifetimes up to 180 seconds. Provider keys and endpoints are checked against certified provider announcements. Its HTTP routes remain restricted to the invited SSH mesh. Record replication does not replicate book/token artifacts. (Newcomer contribution evidence; Implementation and protocol notes)

Sampling versus this run. For uniform sampling without replacement of k chunks from N, with m invalid chunks, P(detect) = 1 - C(N-m,k)/C(N,k). This requires a fixed precommitted bad set and the specified sampling scheme. The current hybrid release uses full assigned-shard replay with two distinct admitted auditor keys per job. It is not a ZKP or proof that a remote signer did the work; rewards, Sybil resistance and malicious-auditor incentives remain open.

4. Hybrid progression and fault tests

The first online baseline pooled role processes within each site's deployment. The next stage used a fresh, separate chain with one container per admitted node. Only public corpus/lineage inputs were reused; the two runs' validator identities were distinct. Later contribution recovery preserved the second chain. (Pooled hybrid evidence; Per-node hybrid evidence)

Measurement	Pooled hybrid baseline	Per-node hybrid baseline
Books / Frags / tokens	8 / 1,438 / 1,657,163	8 / 1,438 / 1,657,163
CPU auditor membership	20	20
Accepted receipts / replays	52 / 2,876	52 / 2,876
Worker release to evidence capture	305.113 s	252.343 s
Cloud CPU allocation	3 vCPU	3 vCPU

These are single executed runs, not a controlled comparative benchmark. The per-node interval includes transfers, replay, coordination, fault exercises and observation cadence. The separate initial seed transfer was 717,987,840 bytes in 121.35 seconds and preceded that interval. All 20 admitted auditors contributed to the per-node run: 16 accepted receipts came from Azure and 36 from local workers. There were 59 attempts, including seven temporary unavailable outcomes that recovered. No expired leases, disputes or rejected transactions were reported for that captured run. (Per-node hybrid evidence)

Deliberate check	Observation
Pause one Oracle process	Remaining three advanced from height 114 to 118
Restart one local Daemon	Same registration and private state retained
Partition Azure/local 2/2	Both halted at 267 for a 15 s stable observation after settling
Reconnect the sites	Chain advanced to 268; all four matched block hash at 296
Probe private application interfaces	TCP ABCI connections refused from another container at each site

The pooled run also halted both halves at 265 and resumed at 266. These bounded experiments support the intended quorum behavior and recovery path. They do not test Byzantine equivocation, prolonged partitions, lost disks, malicious hosts, or tolerance of an entire site outage while maintaining progress. (Pooled hybrid evidence; Per-node hybrid evidence)

5. A newcomer grows the live corpus

The next test modeled a new data contributor with its own container and key, read-only public bootstrap, and two discovery seeds. It announced PROVIDER and TOKENIZER roles, discovered the pinned shared assets, ingested five additional books and submitted a signed contribution manifest. Existing auditors found its endpoint, rebuilt the contribution catalog and replayed the new shards. (Newcomer contribution evidence)

The added works were The Time Machine (35), The War of the Worlds (36), A Princess of Mars (62), Flatland (201) and The Invisible Man (5230). Raw texts were prefetched before the node joined: this was a join/ingestion/audit trial, not a cold download benchmark. The isolated newcomer launched at 01:16:57 Oslo on 14 October.

Observation	Result
Newly contributed Frags / tokens	369 / 404,167
New audit jobs	8, each with two distinct admitted signers
New receipts / checked Frag executions	16 passes / 738 replays
New receipt sites	13 local; 3 Azure; 16 distinct keys total
New failed receipts or remaining leases	0 / 0 at capture
Local audit wall time: min / median / max	18.862 / 25.441 / 26.175 s
Azure audit wall time: min / median / max	112.545 / 112.827 / 113.650 s
Discovery snapshot	All 27 original peers returned the signed live provider record

Audit time includes download, catalog validation and replay. Azure object caches were cold after recovery while local caches were retained. The site timing difference is therefore not an isolated hardware comparison. Historical transport retries are distinct from failures of these eight new jobs.

The incremental Blueprint committed at height 1786, explicitly naming its eight-book parent. Both releases remain immutable. The current union has 13 books, 1,807 Frags, 2,061,330 tokens, 34 completed jobs and 68 accepted receipts. (Newcomer contribution evidence)

A separate keyless reader streamed both released input manifests, checked uniqueness and book-level split separation, and counted 2,059,523 next-token targets. The boundary invariant holds: targets = tokens - chunks. Sequences do not cross source-chunk or split boundaries. This validates an input handoff; no training job was started. New data still has one serving contributor, so automatic artifact replication and provider-loss resilience remain untested.

6. Independent local GPU reconstruction

A fresh diagnostic node ran in its own Podman container on the NVIDIA GeForce RTX 5090 Laptop GPU through WSL. It fetched public inputs through discovered providers, verified fresh Oracle state and the signed lineage, then replayed every released book. Its AUDITOR advertisement is discoverable but grants no lease or consensus-auditor authority. All 29 current peers returned the correct signed, unexpired GPU record. (GPU reconstruction evidence; NVIDIA CUDA references)

Measurement, one full-corpus run	Observed result
Corpus coverage	13 books; 1,807 Frags; 2,061,330 tokens
GPU / NumPy / committed hash comparison	1,807 matches; zero mismatches
Real CUDA launches over corpus	1,807; no CPU fallback
Dense vector bytes reconstructed	6,332,405,760; none persisted
End-to-end fetch and verification	66.577 s
Summed CUDA event kernel interval	211.634 ms
Summed CUDA round-trip interval	6.664 s
Summed NumPy reference lookup	3.323 s
Downloaded application object bytes	172,517,031
Explicit resident GPU buffers	204,786,688 bytes, excluding driver/context overhead

The fixed kernel performs index arithmetic and 32-bit loads/stores. A shared table stays resident, with reusable bounded buffers for one chunk at a time. Each GPU result is compared byte-for-byte with NumPy before the existing verifier checks the vector hash. The current backend lives outside the pinned core source tree, and its script/kernel hashes are recorded in the signed report. No live consensus implementation pin changed.

Distributed-system benefit. The node reconstructed and verified 6.33 GB of dense vectors after fetching 172.5 MB of application objects in 66.6 seconds. That is **36.7 times fewer payload bytes (97.3% less)** than transferring the uncompressed dense vectors alone, including the shared matrix in this fresh-cache fetch. This is a useful bandwidth result for the decentralized design. It is a representation-and-reconstruction benefit shared by CPU and GPU implementations; the test also establishes that a real GPU can reproduce every committed byte.

Timing scope. Small-chunk CUDA round trips totaled 6.664 s versus 3.323 s for NumPy. This does not establish GPU acceleration of the local lookup. Kernel events exclude transfers; CUDA round trips exclude final serialization, while NumPy includes it. The 66.6 s run used local providers and included both GPU and CPU verification. WAN time savings, compressed-vector baselines and warm-cache benefits remain to be measured; they do not negate the observed payload reduction.

Fixtures covered repeated and boundary indices, partial blocks, signed zero, subnormals and the maximum chunk size. Invalid indices and an injected bit flip were rejected. Four focused tests passed, including GPU replay of a signed fixture and rejection of a validly signed but incorrect vector commitment. The node now watches for new Blueprints until the existing pilot deadline.

7. Recovery and operational observations

The move to nine Azure containers exposed a provisioning/ingress failure. Containers could report running while the group stayed Creating without an API-reported public IP. Internal SSH worked; external SSH and HTTPS did not. Management authentication was valid; ordinary restart did not resolve that symptom. Group and container port declarations agreed, and no customer VNet, NSG, route table or managed firewall was attached. (Network recovery evidence)

Closed node archives were hashed and read-verified before sequential disposable network probes. The probes held no node keys or ledger state. Local signers and the real cloud group were stopped during diagnosis, preserving signer fencing. No live validator was cloned to another group.

Selected controlled probe	Outcome
One container, 1 CPU / 1 GB	Public test payload reachable
Eight containers with original allocations	Reachable after 36 s
Nine containers with original explicit limits	Running containers, stalled public provisioning
Nine containers, no explicit limits	Reachable after 98.9 s
Nine, compute limits 3 CPU, auxiliary limits 0.3 / 0.2	Reachable after 37.3 s
Same limits, public 80/443 declarations restored	Port 2222 test payload reachable after 41 s

The real deployment recovered with the smaller gateway/Console CPU limits while retaining 3 vCPU/8 GB total requests. This implicates the tested resource-limit configuration, but does not establish Azure's internal root cause or a general container-count rule. The successful port-declaration probe had no Console listener; real HTTPS was verified separately after deployment. (Network recovery evidence; Azure container-group reference)

All seven cloud node archives restored their original identities and chain history. The chain resumed at 1595 with its old Blueprint; the contribution protocol activated at 1616. The cloud Console obtained a valid HTTPS certificate. Both main Consoles showed fresh certified state and 28 topology/ledger nodes before the GPU addition. The GPU has its own local page and site telemetry; the main Console's fixed node list has not yet been expanded to that diagnostic.

These recovery observations establish same-chain continuity across a controlled deployment change. They do not constitute an unattended long-duration soak test. Infrastructure lifecycle details remain in the operator runbooks; this report focuses on architecture, correctness and experimental evidence.

8. What is established, and what comes next

The platform is useful as an experimental substrate because its claims can be made explicit and replayed. It separates data supply, deterministic processing, audit claims, consensus certification and observational telemetry. That is the foundation needed before studying how visible provenance changes a person's trust in a trained model's output.

Question	Current evidence	Next controlled experiment
Can data be added without rewriting history?	One five-book newcomer, parent-linked release	Multiple simultaneous contributors, duplicate and conflicting proposals
Can independent execution reproduce vectors?	Full CPU coverage plus GPU/CPU/hash equality	Repeat across devices, drivers and batch sizes under fixed conditions
Can the network scale?	20 auditors, 29 current node identities, 13-book corpus	Factorial corpus/concurrency runs; throughput, p95 latency, queues and transferred bytes
Is new data available through churn?	One new serving contributor; discovered by peers	Verified artifact replicas, then remove original provider and measure recovery
Is consensus resilient?	One-process pause and bounded 2/2 partition	Independent hosts/operators; longer partitions, disk loss, equivocation handling
Does provenance improve calibrated trust?	No human study or trained model yet	Controlled provenance visibility and correctness conditions with comprehension checks

For the next performance study, distinguish cold and warm caches, randomize run order, repeat each condition and keep compute allocations fixed. Measure acquisition, tokenization, artifact transfer, replay, certification and publication separately. Record failures and retries alongside successful throughput. Choose acceptance thresholds before observing comparative results. The present timings are useful engineering observations, not confidence intervals or scaling laws.

The compact transfer result supports reconstructing dense vectors near their consumer. The GPU result also motivates batching or retaining vectors on device when later compute uses them; local lookup acceleration is a separate question. Training nodes and a complete trained-weight lineage remain subsequent stages. The imported model/weight anchors and input inspection mechanisms are groundwork, not a DAWL-trained model or proof of upstream training data. (GPU reconstruction evidence; Implementation and protocol notes)

Admission certificates, contribution reputation, rewards, slashing and dynamic validator governance remain deferred. Signed claims alone do not solve collusion, Sybil identities, fabricated approvals or eclipse attacks. Role separation and container count must not be presented as independent economic actors.

9. Evidence, pins and reproducibility

The companion ZIP contains this report, editable documentation, focused GPU implementation/tests, standalone architecture sheets and selected public evidence. Its SHA-256 manifest and verification script check package integrity and the GPU report signature. They do not replace a full CometBFT certificate verifier. The pilot's Go verifier checked the captured state certificates against the pinned genesis at execution time. Historic discovery records are checked at their recorded observation time, not treated as live advertisements. (Per-node hybrid evidence; Newcomer contribution evidence; GPU reconstruction evidence)

Evidence set	Archive path and purpose
Pooled hybrid evidence	evidence/pooled/ - first hybrid completion and partition snapshots
Per-node hybrid evidence	evidence/per-node/ - completion, audit analysis, faults, convergence and certificate
Newcomer contribution evidence	evidence/contribution/ - upgrade, newcomer, release, signed discovery and input check
GPU reconstruction evidence	evidence/gpu/ - signed report, runtime, input state, later certificate and discovery
Network recovery evidence	evidence/recovery/ - observed ingress controls and final deployment checks
Implementation and protocol notes	implementation/ and docs/ - focused reproducible code and protocol documentation

The full DAWL repository is required for live replay. Reacquire the exact hash-pinned public source/tokenizer/embedding assets and public bootstrap; use fresh node identities. Never copy a live validator's keys or signing database. The GPU runbook documents the WSL-specific build, launch, route and export steps. Private keys, storage credentials, signed-URL secrets, private state archives and raw corpus/model binaries are excluded from this handoff.

Chain: `hypertrix-dawlh1013185918`

```
Genesis: 8443e106c542c1e920e51862b9763300bc2079871f9d03f1cc3c46ded5de1871
Network: c2a69886ef30d988418d8dd9d130cdec15b8b5141c3ad9b14e02bd1dd6b1c888
Blueprint: fd8760e4f3878a1bcb79645f42f0f55bcbbaab74f20d9aa2cfe53ac005743950
GPU report: 26145a140ce3d60a3f9b26aec997b3b4c3b7f7f256253657b38e4c7f2339efd6
```

Primary technical references

Microsoft: Container groups in Azure Container Instances. Shared group resources/networking and public port declarations; accessed 14 October 2026. https://learn.microsoft.com/en-us/azure/container-instances/container-instances-container-groups

CometBFT specification and verifier as used by the repository's pinned Go module: consensus/go.mod and consensus/main.go. The four-validator experiment does not claim deployment of independently administered validators.

NVIDIA: CUDA Driver API: execution control and event management; CUDA on WSL. Used for the fixed Gather kernel and timing, not as evidence for measured results. https://docs.nvidia.com/cuda/cuda-driver-api/cuda_driver_api/group__CUDA__EXEC.html https://docs.nvidia.com/cuda/cuda-driver-api/cuda_driver_api/group__CUDA__EVENT.html https://docs.nvidia.com/cuda/wsl-user-guide/contents.html